

QUESTIONNAIRE

1) Loading and first look

1. Read the CSV.
2. Print shape, column names, and dtypes.
3. Show the first 10 rows. Which columns are numeric, and which are text?
4. Check for missing values across all columns.
5. How many unique metro lines are there? List them with their station counts

2) Cleaning

6. Opened (Year) is stored as text like 08-03-2019 — convert it to a proper datetime column.
7. ID (Station ID) looks like it should be a unique key. Check `df['ID (Station ID)']. duplicated(). sum()`
8. Are there any duplicate Station Names? Look them up — are they genuinely the same station or two different stations that happen to share a name?

3) Filtering and sorting

9. Show all stations on the Blue line.
10. Which stations are Underground vs Elevated vs At-Grade? Count each.
11. Find every station more than 40 km from its line's first station.
12. Sort stations by Opened (Year) — which was the very first station to open, and which is the most recent?

4) Grouping and aggregation

13. Average distance from first station, grouped by Metro Line — which line is longest end-to-end?
14. Count of stations per Metro Line, sorted descending.
15. Group by Layout and Metro Line together — which lines are mostly underground vs mostly elevated?
16. How many stations opened each year? Which year had the most openings?

5) Feature creation

17. Extract just the year from Opened (Year) into a new integer column.

6) Visualisation with Seaborn/Matplotlib

18. Bar plot of station count per Metro Line
19. Count plot of Layout (Elevated/Underground/At-Grade).
20. Histogram of Dist. From First Station(km) — what shape is the distribution?